series shows a discernible break around the launch of ChatGPT (vertical dashed line at 2023Q1), indicating that the compositional structure of the LinkedIn sample is essentially unchanged across the pre- and post-GenAI periods.

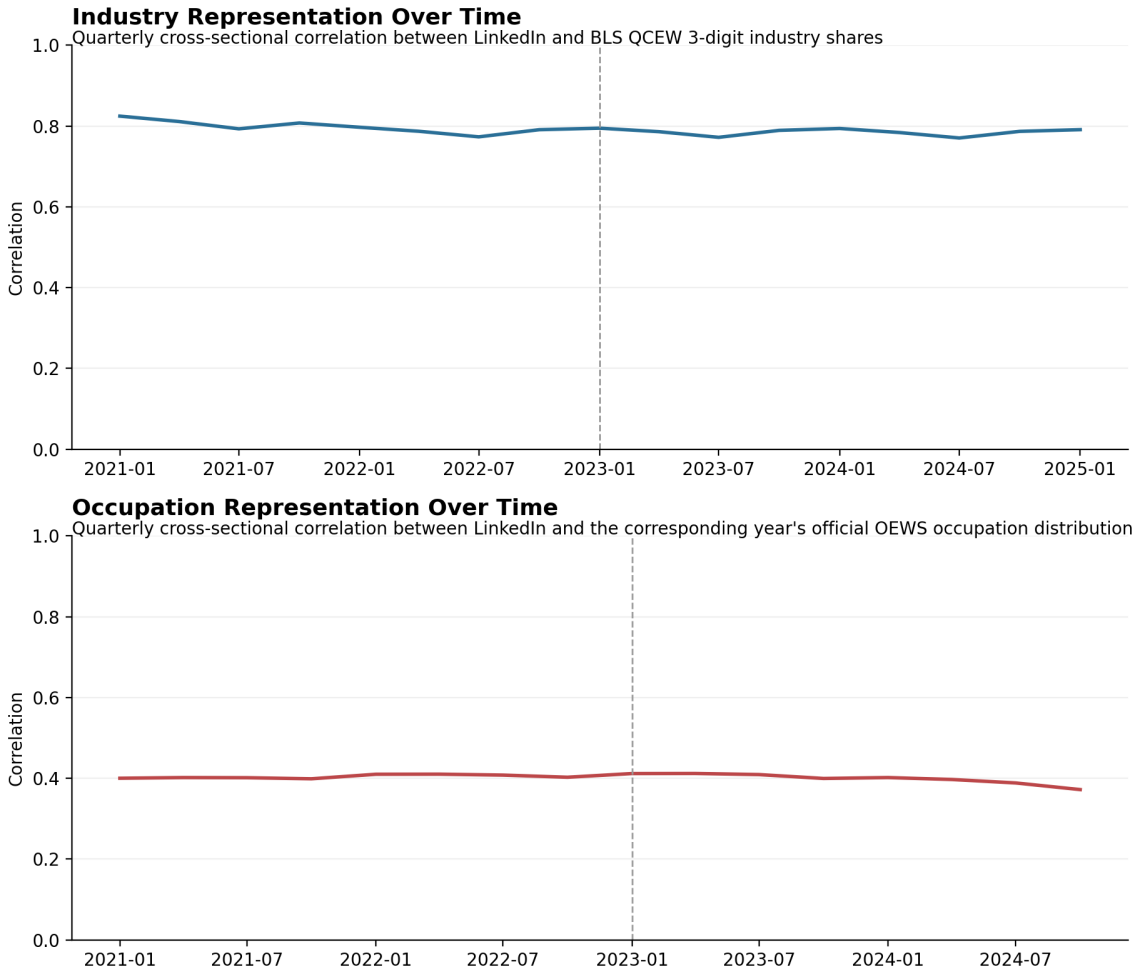


Figure A.4: Stability of Industry and Occupation Representation Over Time

Figure A.5 repeats the exercise for the age dimension. The top panel plots the quarterly cross-sectional correlation between LinkedIn junior sector shares and the corresponding QWI under-30 sector shares; the bottom panel does the same for senior versus 30+ shares. The junior correlation is stable around 0.44 throughout the sample, and the senior correlation is stable around 0.60, with neither series showing a break around the GenAI launch. The relative representation of junior and senior workers across sectors in the LinkedIn data therefore evolves in line with the official QWI age structure, supporting the use of LinkedIn-based seniority composition to identify changes in junior and senior employ-

ment around GenAI adoption.

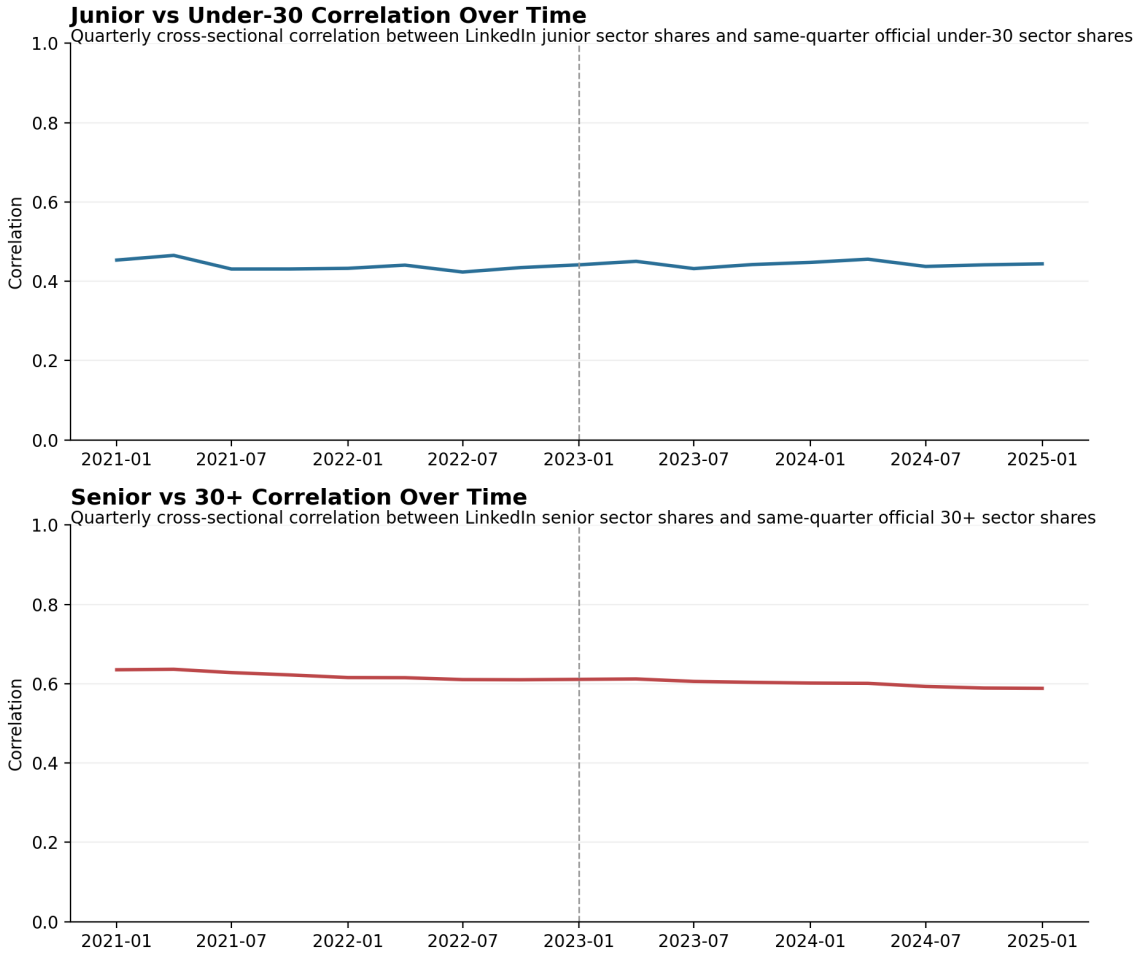


Figure A.5: Stability of Age-Proxy Representation Over Time

A.2 Validation of the Seniority Classification

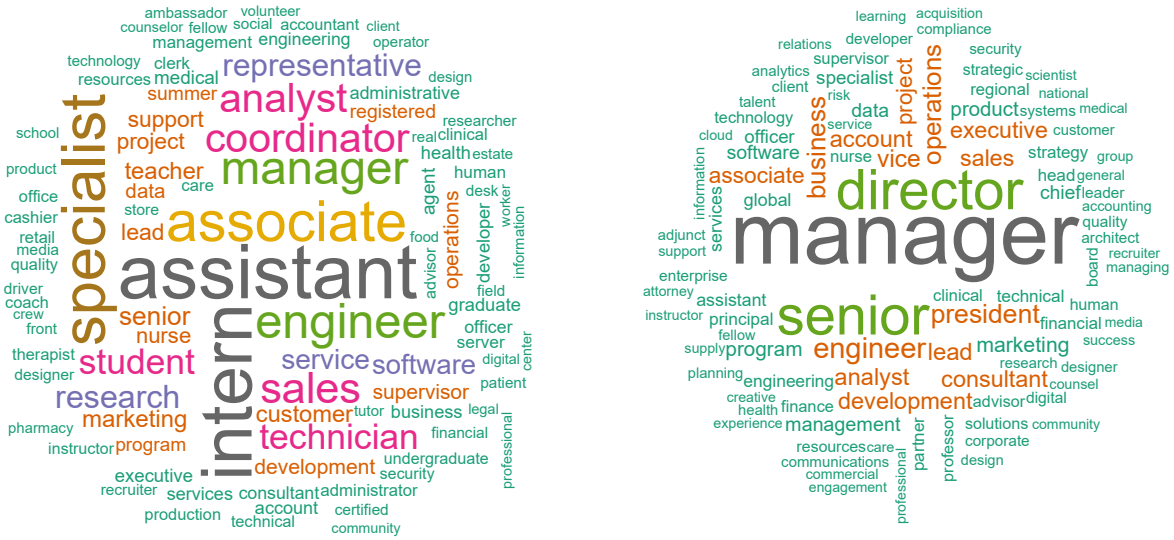
We conduct two complementary exercises to evaluate the seniority classification produced by Revelio and to verify its stability over the GenAI diffusion period. First, we analyze job title keywords for junior and senior roles. Second, we examine the average experience levels within each seniority group. In both exercises, we study patterns over time and across adoption status.

A.2.1 Job Title Keywords by Seniority

What junior and senior positions look like. We first examine the distribution of keywords in raw job titles, separately for junior and senior workers. Figure A.6 presents word clouds summarizing the most frequently occurring keywords in each seniority group across the full sample. The classification is meaningful: junior positions are dominated by terms such as *assistant*, *intern*, *specialist*, *associate*, *technician*, and *analyst*, while senior positions are concentrated around *manager*, *director*, *senior*, *consultant*, *president* and *lead*. These patterns confirm that Revelio's seniority variable aligns well with conventional titling and motivate the keyword lists used in the exercises below.

Stability of keyword distributions across 2023. We next compare the distribution of keyword frequencies in positions that started before 2023 to those that started in 2023 or later, pooled across all firms. Figure A.7 shows that the relative prominence of keywords is stable over time within each seniority group. Junior positions continue to be dominated by the same keywords before and after 2023, and senior positions by theirs. The main exception is *intern*, whose share increases after 2023 relative to the earlier period. This stability argues against a systematic shift in titling conventions or classification methodology around the diffusion of GenAI.

Differential evolution at adopting vs. non-adopting firms. The previous exercises pool all firms together. To directly address the concern that *adopting* firms may be reclassifying positions differentially, we track the evolution of keyword shares over time, separately at adopting and non-adopting firms. For each firm-quarter cell, we compute two shares: the share of junior-classified positions whose raw title contains any junior-leaning



(a) Juniors (Seniority 1–2)

(b) Seniors (Seniority 3+)

Figure A.6: Most Common Job-Title Keywords by Seniority Group

Notes: Word clouds summarizing the most frequently occurring keywords in raw job titles, computed separately for junior-classified positions (seniority 1–2) and senior-classified positions (seniority 3+), pooled across positions that started between 2020Q1 and 2025Q1. Larger font size indicates higher frequency.

keyword (*assistant*, *intern*, *junior*, *coordinator*, *specialist*, *technician*, *entry*), and the share of senior-classified positions whose raw title contains any senior-leaning keyword (*manager*, *director*, *senior*, *lead*, *principal*, *consultant*, *supervisor*, *executive*, *head*, *chief*). Figure A.8 plots these two shares quarterly, separately for adopting and non-adopting firms. The four series evolve in close parallel. Roughly a quarter of junior positions contain a junior-leaning keyword, and roughly two-thirds of senior positions contain a senior-leaning keyword, at both adopting and non-adopting firms throughout the sample. Pre-existing level gaps between adopters and non-adopters are stable around the ChatGPT launch.

Event study. To formally test for differential trends, we estimate, separately for juniors and seniors, the specification

$$\text{Share}^{\text{match}}_{f,t} = \alpha_f + \alpha_t + \sum_{k \neq 2022\text{Q}4} \beta_k \cdot \mathbf{1}\{t = k\} \cdot \text{Adopter}_f + \varepsilon_{f,t},$$